

Modeling Climate Anomalies
with Statistical Analysis

Today's goals



Module 1

Introduction to Python for Data Analysis

Course Introduction

- ✓ Course Updates and Accessibility Support
Reading • 1 min
- ✓ Earn Academic Credit for your Work!
Reading • 10 min
- ✓ Course Support
Reading • 10 min
- Introduction to the Course
Video • 2 min
- Meet the Instructor
Video • 1 min
- ✓ Introduction to Pandas for Data Loading and
Exploration
Video • 25 min
- ✓ Pandas
Reading • 15 min
- ✓ Course Assignments with Jupyter
Reading • 10 min

Visualizing Data with Python

- ✓ Introduction to Matplotlib and Seaborn for Data
Visualization
Video • 25 min

Module Assignment

- ✓ Introduction to Pandas and Matplotlib
Programming Assignment • Grade: 100%
- ✓ Pandas and Matplotlib
Graded Assignment • Grade: 93.33%

Module 2

Collecting Climate Data

Data Collection and APIs

- ✓ Introduction to APIs
Video • 9 min
- ✓ APIs, JSON, and Python Requests
Reading • 15 min
- API Hands-On (Lab Activity)
Reading • 30 min
- API Hands-On
Graded Assignment • 15 min

Data Sources: NOAA

- Introduction to NOAA Website for Data
Collection
Video • 9 min
- NOAA API and Available Data
Reading • 10 min

Data Sources: USGS

Introduction to USGS Website and Available Data

APIs, JSON, and Python Requests



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2x



Application Programming Interfaces (APIs) play a crucial role in enabling different software systems to communicate with each other. APIs act as intermediaries, allowing applications to request and exchange data seamlessly. This functionality is fundamental for integrating various services, such as retrieving weather data, accessing social media feeds, or interacting with cloud services.

APIs typically provide data in a structured format known as JSON (JavaScript Object Notation). JSON is a lightweight data-interchange format that is easy for humans to read and write, and easy for machines to parse and generate. Its simplicity and flexibility make it the preferred format for many APIs, allowing developers to work with data in a consistent manner.

To interact with APIs in Python, the requests library is commonly used. This powerful library simplifies the process of sending HTTP requests and handling responses. With requests, you can easily perform various HTTP methods to communicate with APIs. The library also provides mechanisms for handling response status codes, errors, and extracting data from JSON responses.

To learn more about Python Requests, please see the following user guides:

[Requests Quickstart](#)  (external link; opens in a new tab)

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